

Retail Sales & Customer Analytics Using Python

IBM Data Analytics Internship — Project Report

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Project Type: Data Analytics / Business Intelligence

1. Executive Summary

This project demonstrates a practical data analytics workflow using a synthetic retail sales dataset. The analysis covers data cleaning, exploratory data analysis, KPI calculation, trend analysis, category performance, product performance and customer-level analysis. The objective is to convert raw sales data into clear business insights that can support inventory, sales and customer-management decisions.

2. Objectives

- Analyze revenue and profit trends.
- Compare product categories.
- Identify high-performing products.
- Understand customer contribution.
- Create business-focused visualizations.
- Provide actionable recommendations.

3. Dataset

The dataset contains 1,200 synthetic retail sales records covering January–December 2025. Fields include Date, Customer_ID, Region, Category, Product, Quantity, Unit_Price, Discount, Revenue, Cost and Profit. The dataset is synthetic and was created specifically for this portfolio/internship demonstration.

4. Tools & Technologies

Python, Pandas, NumPy, Matplotlib, CSV and Jupyter/VS Code.

5. KPI Summary

Metric	Result
Total Revenue	1,142,749.16
Total Profit	309,118.54
Profit Margin	27.05%
Sales Records	1,200
Unique Customers	180

6. Key Findings

The highest-revenue category in the generated dataset is **Grocery**. The highest-revenue product is **Pulses**. The strongest revenue month is **2025-10**. These results illustrate how category, product and time-based analysis can be used to prioritize inventory and sales attention.

7. Visual Analysis

The project produces four visualizations: monthly revenue trend, category revenue, top products and quantity versus profit.

8. Recommendations

1. Maintain sufficient inventory for high-revenue products.
2. Review low-performing categories before increasing stock.
3. Use customer-level revenue analysis for targeted offers.
4. Monitor discounts to protect profit margins.
5. Track monthly KPIs consistently to identify changes early.

9. Conclusion

The project shows how a structured analytics workflow can turn raw transaction data into useful business information. It demonstrates core skills expected in an entry-level data analytics role: data cleaning, aggregation, KPI analysis, visualization and business communication.

Note: All numerical results are based on the synthetic dataset included with this project.